PASSIVE RECONNAISSANCE

TryHackMe Learning & Practical Report

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Program	IqraSec Academy, WAPT Internship (Cohort 02), Phase 1
Task	2. Recon & Enumeration
Platform	TryHackMe
Room	Passive Reconnaissance (https://tryhackme.com/room/passiverecon)
Focus	passive recon, DNS, whois, nslookup, dig, DNSDumpster, Shodan
Completion	100% (all 7 tasks)
Date	October 2026

1. Objective

Understand passive reconnaissance: how to collect information about a target (domain, DNS records, servers) without interacting with it directly.

2. Concept / Understanding

- **Passive recon:** collecting information without touching the target systems, using public sources. The target does not see our activity. It is like watching from far away.
- **Active recon:** interacting directly with the target (for example scanning it). The activity can be seen and logged by the target's defenses. It is like trying to open the lock yourself.
- **DNS names:** TLD is the top-level domain (.com). Domain name is the core name (tryhackme). Subdomain is the prefix (www). FQDN is the full name together (www.tryhackme.com).
- **DNS resolution steps:** check the local cache/files, ask the recursive DNS server (ISP), then the root server, then the TLD server, and finally the authoritative name server that returns the IP.
- **whois:** shows the domain registration data: registrar, creation and expiry dates, name servers.
- **nslookup and dig:** query DNS records of a domain from the command line.
- **DNSDumpster:** an online tool that lists subdomains and DNS records of a domain without touching the target.
- **Shodan.io:** a search engine for internet-connected devices. It shows IP addresses, open ports and services.
- **Why it matters:** the information (subdomains, servers, mail servers) helps to understand the attack surface before testing, and passive recon does not trigger alerts.

DNS record types:

Record	Use
A	maps a domain to an IPv4 address
AAAA	maps a domain to an IPv6 address
CNAME	alias that points to another domain name
MX	mail servers of the domain
TXT	free text, used for SPF/DMARC and domain verification

3. Practical Work

I finished all 7 tasks of the room (Introduction, Passive Versus Active Recon, Whois, nslookup and dig, DNSDumpster, Shodan.io, Summary). In Task 4, I opened a terminal on the AttackBox and

used DNS lookup commands to check the **TXT records of thmlabs.com**. The flag was inside the TXT record, and I submitted it as the answer.

4. Tools Used

- TryHackMe AttackBox (terminal)
- nslookup / dig (DNS records)
- whois (domain registration data)
- DNSDumpster (subdomains and DNS records)
- Shodan.io (internet-connected devices)

5. Evidence

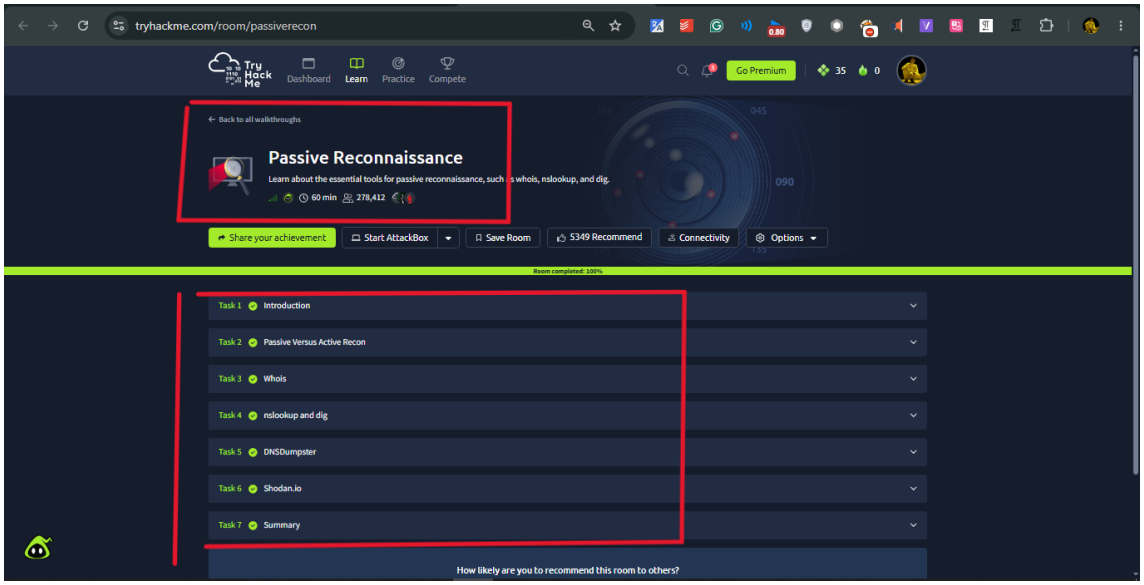


Figure 1: Passive Reconnaissance room page, 100% completed, all 7 tasks done.

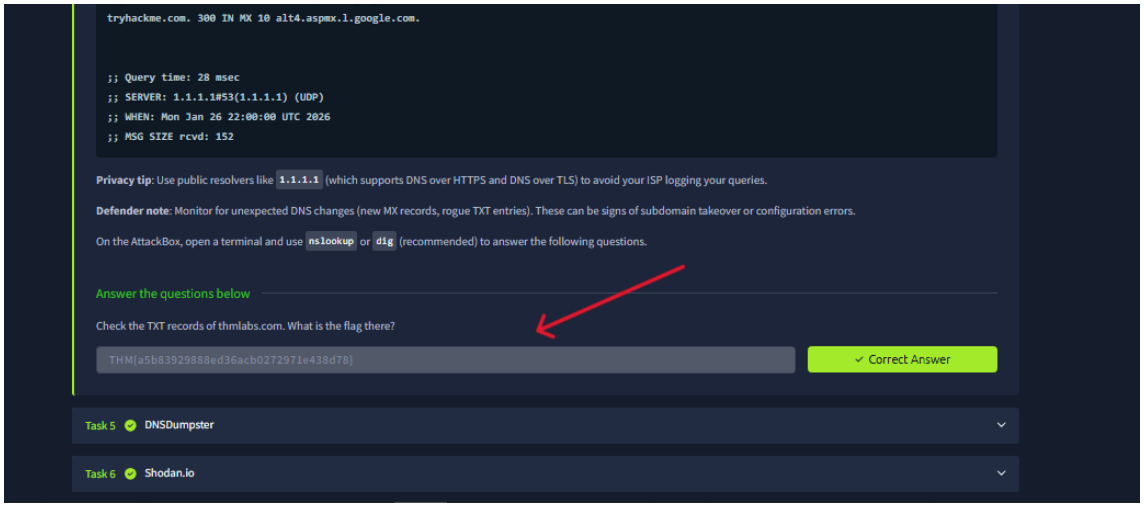


Figure 2: Task 4, the answer for the TXT records of thmlabs.com (marked with the arrow), accepted as correct.

6. Result

Completed the room 100%. I found the flag in the TXT record of thmlabs.com using DNS lookup commands, without any direct interaction with the target.

7. Key Learnings

- Passive recon collects information without touching the target, while active recon interacts with it and can be detected.
- How DNS turns a domain name into an IP address, step by step.
- DNS record types (A, MX, TXT) show useful information about a target, like mail servers and verification data.
- whois, DNSDumpster and Shodan.io give a lot of public information about a target without sending traffic to it.